Student Research Project Proposal

Image Augmentation for Small Datasets

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1 Problem Setting

Deep image classification systems are often developed and evaluated in settings with abundant labeled data, exemplified by large-scale benchmarks such as ImageNet. However, in many practical domains, labeled data are limited due to the cost of annotation, the need for expert labeling, privacy constraints, or simply limited data collection opportunities. When the number of labeled samples is small, modern high-capacity models can overfit or rely on spurious shortcuts, leading to degraded robustness and poor generalization to unseen data.

Data augmentation is a common strategy to reduce overfitting by increasing training diversity through label-preserving transformations and sample-mixing techniques. Traditional methods (e.g., geometric and photometric transformations) and mix-based approaches such as MixUp and CutMix have shown strong performance on standard benchmarks, while AugMix improves robustness through stochastic augmentation and prediction consistency. However, augmentation is only effective when transformations remain semantically valid for the task, as overly strong transformations may introduce label noise and degrade performance, especially in small or underrepresented datasets.

This project studies image augmentation for small datasets with a focus on systematic evaluation and method design. We will compare common augmentation baselines (e.g., MixUp, CutMix, AugMix) against no augmentation under controlled low-data regimes created by subsampling standard image datasets. We will evaluate augmentation effects across a convolutional architecture (ResNet) and a transformer-based architecture (Vision Transformer), whose training behavior is known to be sensitive to data scale and training recipe choices. Beyond reproducing and comparing existing methods, we aim to design and evaluate a lightweight augmentation variant tailored to small-data settings.

The central research question is:

> Which image augmentation strategies improve classification performance most reliably in low-data settings across both CNNs and vision transformers, and can a small-data-targeted augmentation variant outperform or complement standard augmentation baselines?

2 State-of-the-Art

Data augmentation improves generalization in image classification by increasing training-set diversity without collecting additional labels. Augmentation methods are commonly grouped into (i) label-preserving transformations applied to individual images, and (ii) multi-sample approaches (mixing or composition) that synthesize new training examples from multiple images.

Classical label-preserving transforms. Standard pipelines apply lightweight geometric and photometric transformations such as random crops, flips, small rotations, and color jitter. Their effectiveness depends on the extent to which the induced invariances are valid for a given task: if transformations alter class-relevant semantics, they can introduce effective label noise. This consideration becomes particularly important in low-data regimes, where each sample carries substantial information about the underlying data distribution [1].

Masking and deletion-based regularization. Cutout improves robustness by randomly masking image regions during training and has shown consistent gains on common benchmarks [2]. Closely related approaches, such as Random Erasing, similarly simulate occlusions by removing random rectangular regions [3]. These methods can encourage models to rely on broader evidence rather than a single discriminative patch, but may also remove informative regions when data are scarce.

Sample-mixing methods. MixUp trains on convex combinations of images and labels, encouraging smoother decision boundaries between examples [4]. CutMix replaces an image patch with a patch from another image and mixes labels proportionally to the replaced area, aiming to retain more local information than global blending [5]. While strong on standard benchmarks, mixing strength and realism can critically affect performance, motivating careful evaluation under low-data constraints.

Stochastic composition and policy-based augmentation. AugMix composes multiple stochastic augmentations and enforces consistency, improving robustness while maintaining clean accuracy [6]. Policy-based methods such as AutoAugment search for augmentation policies [7], while RandAugment reduces this complexity by parameterizing policy strength and avoiding an expensive separate search phase [8]. Recent work (e.g., TrivialAugment) further shows that competitive policy-style augmentation can be achieved with minimal tuning overhead [9].

Architecture interaction and research gap. Augmentation effectiveness depends not only on datasets but also on model inductive bias. ResNets incorporate convolutional priors, whereas Vision Transformers were originally shown to benefit strongly from large-scale pretraining and often require careful augmentation/regularization when trained on smaller datasets [10, 11]. Despite many proposed methods, there is limited systematic understanding of which augmentations are most reliable in *small-data regimes* under fair protocols that vary (i) the number of labeled samples, and (ii) model family (CNN vs. ViT). This project addresses this gap via controlled subsampling experiments and a targeted small-data augmentation variant.

3 Data Foundation

The project is based on publicly available benchmark datasets and evaluates augmentation methods under controlled low-data conditions created by subsampling the training set.

Datasets. We use **CIFAR-10** (10 classes) and **CIFAR-100** (100 classes) as primary benchmarks. Both datasets are standardized, balanced, and computationally efficient, enabling repeated experiments and systematic ablations. While CIFAR-10 represents a relatively simple classification setting, CIFAR-100 introduces higher class granularity, making it more challenging under limited data. This combination allows us to study augmentation effects across different levels of task complexity.

Low-data regime construction. To simulate small-data scenarios, we create stratified k-shot subsets of the training set with

$$k \in \{5, 10, 20, 50, 100\}$$

samples per class. For each (dataset, k) configuration, we generate multiple random subset seeds to reduce sampling bias. Validation and test splits remain fixed across all augmentation methods and architectures.

4 Research Idea

Concept. We propose **Similarity-Guided Mixing**, a small-data-oriented modification of sample-mixing augmentations. Instead of selecting MixUp or CutMix pairing partners uniformly at random, Similarity-Guided Mixing chooses partners that are *semantically similar* to the current sample.

Motivation and Intuition. In low-k regimes, random MixUp/CutMix can generate ambiguous samples by combining images from unrelated classes or visually dissimilar modes, potentially diluting the already scarce class-defining evidence. By pairing each image with a nearest-neighbor in feature space, Similarity-Guided Mixing aims to produce mixed examples that remain closer to the data manifold, thereby improving regularization while preserving class-relevant structure. This is expected to reduce harmful label ambiguity and improve stability.

Hypotheses.

- In small-data regimes, Similarity-Guided Mixing strategy could achieve higher test accuracy than standard MixUp/CutMix at the same training budget.
- Similarity-Guided Mixing reduces performance variance across subset seeds, indicating improved robustness to sampling noise.
- The benefit of similarity-guided pairing differs between ResNet and ViT, reflecting architecture-dependent interactions with augmentation.

Method sketch (high-level). The key idea is to *guide the choice of the mixing partner* in MixUp/CutMix using similarity, rather than selecting partners uniformly at random. We implement this in two stages:

Stage 1: Similarity-guided partner selection (offline). We compute a fixed feature representation for each training image using a frozen pretrained encoder $g(\cdot)$ (no fine-tuning), obtaining embeddings $z_i = g(x_i)$. We then construct for each image x_i a K-nearest-neighbor set $\mathcal{N}_K(i)$ in embedding space using cosine similarity. During training, the mixing partner j for sample i will be drawn from $\mathcal{N}_K(i)$ (uniformly or weighted by similarity). We consider two pairing variants: (i) *class-aware* neighbors, where neighbors are searched only among samples with the same label ($y_j = y_i$) to reduce label ambiguity, and (ii) *class-agnostic* neighbors, where neighbors may come from any class to increase diversity; this choice will be evaluated in ablations.

Stage 2: Guided mixing during training (online). During training, when a sample x_i is selected, we choose its mixing partner x_j by sampling from $\mathcal{N}_K(i)$ (e.g., uniformly or weighted by similarity), instead of sampling j from the entire dataset. After selecting (x_i, x_j) via this guided procedure, we apply the standard mixing operators:

- **SimMixUp:** apply MixUp to the selected pair with coefficient λ: $(\tilde{x}, \tilde{y}) = (\lambda x_i + (1-\lambda)x_j,\ \lambda y_i + (1-\lambda)y_j)$.
- **SimCutMix:** apply CutMix to the selected pair by pasting a random patch from x_j into x_i and mixing labels proportionally to the pasted area (as in CutMix).

Optionally, we use a **warm-up schedule** that increases the probability of applying mixing over the first epochs, reducing the risk of overly strong regularization at the start of training.

Validation plan (high-level). We evaluate Similarity-Guided Mixing on CIFAR-10/100 under controlled k-shot subsets ($k \in \{5, 10, 20, 50, 100\}$) with multiple subset seeds. For each (dataset, k, seed), we train both a ResNet and a ViT and compare: No Augmentation, MixUp, CutMix, AugMix and our variant. Ablations include neighbor count K, embedding choice, and the impact of warm-up. Failure cases will be studied via confusion matrices and qualitative inspection of mixed samples where Similarity-Guided Mixing helps or hurts.

5 Tangible Outcomes

A key outcome of this project will be the preparation and submission of a workshop paper presenting the core ideas, methodology, and preliminary results of the study. The work will be submitted to a relevant academic workshop or conference to contribute to ongoing research discussions and obtain feedback from the scholarly community.

6 Work Plan

Time Period	**Planned Activities**
December 2025 – March 2026	Team formation, refinement of the research idea, and preparation of the project proposal.
April 2026 – May 2026	Development of a baseline approach and formulation of the initial methodological framework.
June 2026 – September 2026	Design and implementation of the proposed model or software, incorporating the team's original research ideas.
October 2026 – November 2026	Systematic evaluation of the developed model, including performance analysis, feedback-driven improvements, and ablation studies.
December 2026	Final presentation of the project results.

The implementation of the project will be carried out using the Python programming language, leveraging its extensive ecosystem of machine learning libraries and packages. Furthermore, the project intends to build upon existing software frameworks while integrating the research concepts previously outlined in Section 4.

7 Resources

The project will use the resources mentioned below:

- Google Colab virtual machine powered with GPU/TPU

- ISMLL GPU cluster

8 Team

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